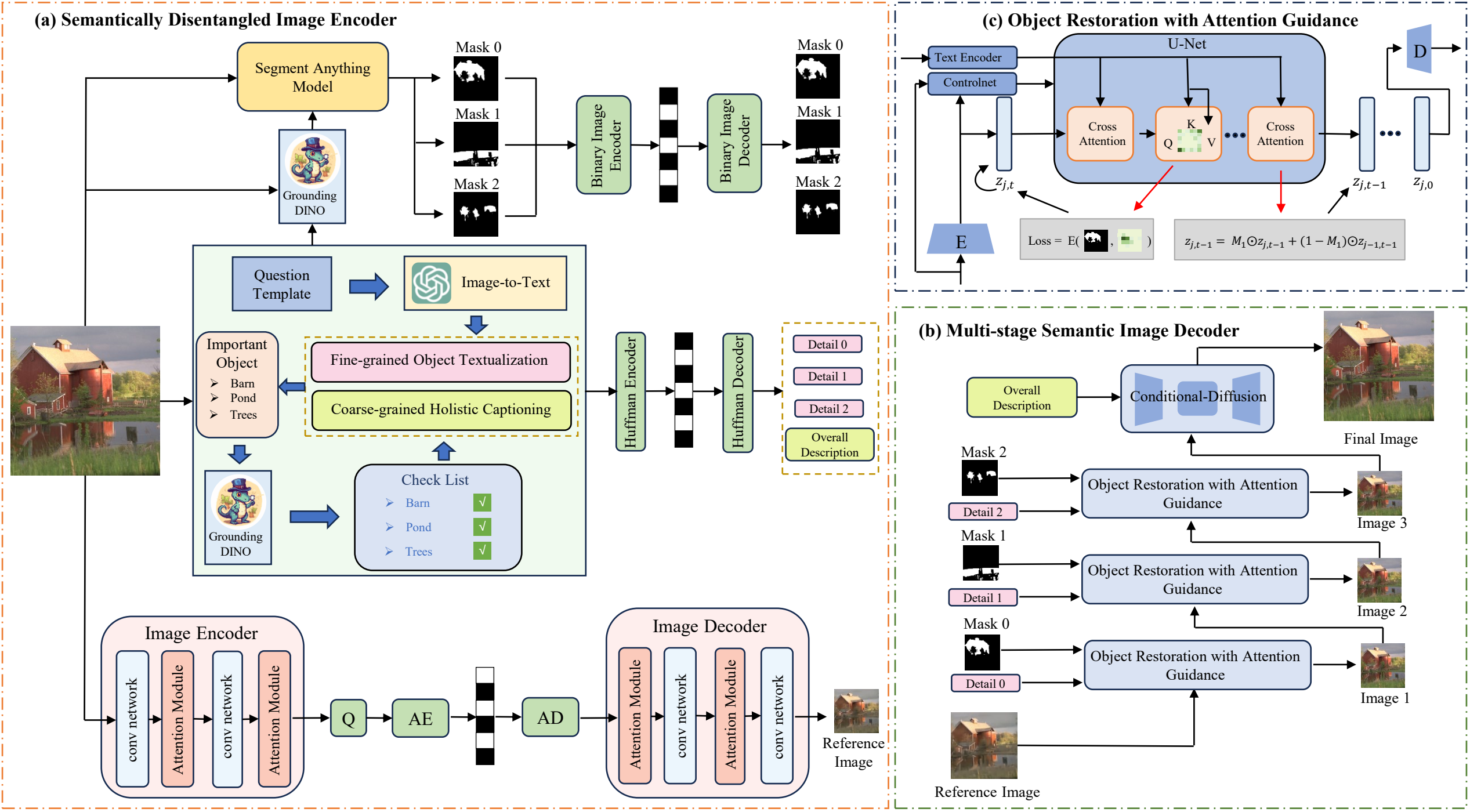

(a) Semantically Disentangled Image Encoder
Segment Anything Model
Grounding DINO
Mask 0
Mask 1
Mask 2
Binary Image Encoder
Binary Image Decoder
Question Template
Image-to-Text
Important Object
Barn
Pond
Trees
Fine-grained Object Textualization
Coarse-grained Holistic Captioning
Check List
Barn √
Pond √
Trees √
Huffman Encoder
Huffman Decoder
Detail 0
Detail 1
Detail 2
Overall Description
Image Encoder
conv network
Attention Module
conv network
Attention Module
Q
AE
AD
Image Decoder
Attention Module
conv network
Attention Module
conv network
Reference Image
(c) Object Restoration with Attention Guidance
Text Encoder
Controlnet
U-Net
Cross Attention
K
Q
V
Cross Attention
D
E
Loss = E(,)
$z_{j,t}$
$z_{j,t-1}$
$z_{j,0}$
$z_{j,t-1} = M_1 \odot z_{j,t-1} + (1 - M_1) \odot z_{j-1,t-1}$
(b) Multi-stage Semantic Image Decoder
Overall Description
Conditional-Diffusion
Final Image
Mask 2
Detail 2
Object Restoration with Attention Guidance
Image 3
Mask 1
Detail 1
Object Restoration with Attention Guidance
Image 2
Mask 0
Detail 0
Object Restoration with Attention Guidance
Image 1
Reference Image